

①

ITL-24655

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Solution of (a):

$$x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$$

1. Start with first two congruences:

Let $x = 3a + 1$, substitute into the second congruence

$$3a + 1 \equiv 2 \pmod{5}$$

$$\Rightarrow 3a \equiv 1 \pmod{5}$$

The inverse of 3 modulo 5 is 2 so:

$$a \equiv 2 \pmod{5}$$

$$\Rightarrow a = 5b + 2$$

$$\text{Then } x = 3(5b + 2) + 1 = 15b + 7$$

2. Now use third congruence

$$15b + 7 \equiv 3 \pmod{7}$$

Since $15 \equiv 1 \pmod{7}$ & $7 \equiv 0 \pmod{7}$ then simplifies to

$$b \equiv 3 \pmod{7} \Rightarrow b = 7c + 3$$

Then,

$$x = 15(7c + 3) + 7 = 105c + 52$$

$$x \equiv 52 \pmod{105} \Rightarrow 52$$

Ans.

②

Solution of 6:

$$x \equiv 5 \pmod{11}, x \equiv 14 \pmod{29}, x \equiv 15 \pmod{31}$$

1. Combine the first two congruence

$x = 11a + 5$. Substitute into the second congruence

$$11a + 5 \equiv 14 \pmod{29}$$

$$11a \equiv 9 \pmod{29}$$

The inverse of 11 module 29.

That means we want

$$11x \equiv 1 \pmod{29}$$

$$\therefore 11 \cdot 8 \equiv 1 \pmod{29}$$

So the inverse is 8. Multiply both sides:

$$a \equiv 9 \times 8 \equiv 72 \equiv 14 \pmod{29}$$

$$a = 29b + 14$$

Substitute back

$$x = 11(29b + 14) + 5 = 319b + 159$$

So,

$$x \equiv 159 \pmod{319}$$

2. Combine with third congruence

$$319b + 159 \equiv 15 \pmod{31}$$

$$319 \equiv 9 \pmod{31} \text{ \& } 159 \equiv 4 \pmod{31}$$

$$\text{So: } 9b + 4 \equiv 15 \pmod{31} \Rightarrow 9b \equiv 11 \pmod{31}$$

(3)

Find the inverse of 9 module 31:

$$9 \times 7 = 63 \equiv 1 \pmod{31}$$

So the inverse is 7. Multiply both sides:

$$b = 11 \times 7 = 77 \equiv 15 \pmod{31} \Rightarrow b = 31c + 15$$

Substitute back

$$x = 319(31c + 15) + 159$$

$$= 9889c + 4944$$

$$\therefore x = 4944 \pmod{9889}$$

$$\Rightarrow 4944$$

Ans.

Solution of (C):

$$x \equiv 5 \pmod{6}, x \equiv 4 \pmod{11}, x \equiv 3 \pmod{12}$$

1. Combine the first two congruence

$x = 6a + 5$ Substitute into the second congruence:

$$6a + 5 = 4 \pmod{11}$$

$$6a = -a \equiv 10 \pmod{11}$$

The inverse of 6 module 11 is 2

Since, $6 \times 2 = 12 \equiv 1 \pmod{11}$

$$a = 2 \times 10 = 20 \equiv 9 \pmod{11}$$

$$\Rightarrow a = 11b + 9$$

Then $x = 6(11b + 9) + 5 = 66b + 59$

2. Now use the 3rd congruence

$$66b + 59 \equiv 3 \pmod{17}$$

Reduce mod 17:

$$66 \equiv 15 \pmod{17}, 59 \equiv 8 \pmod{17}$$

$$\text{So, } 15b + 8 \equiv 3 \pmod{17} \Rightarrow 15b \equiv -5 \pmod{17} \\ \equiv 12 \pmod{17}$$

The inverse of 15 mod 17 is 8

(Since $15 \times 8 = 120$)

$$\therefore 15 \times 8 = 120 \equiv 1 \pmod{17}$$

$$b = 8 \times 12 = 96 \equiv 11 \pmod{17}$$

$$\Rightarrow b = 17c + 11$$

Then,

$$x = 66(17c + 11) + 59 \\ = 1122c + 785$$

$$\therefore x \equiv 785 \pmod{1122} \\ \Rightarrow 785$$

Ans